# OpenSourceRail v0.5

Ferrite PM maglev — primary design reader edition

Fixed magnetic modules · compact elevated guidebeams · Rust controls · local manufacture

Provisional digital studies. Complete vehicle, civil and software qualification remains open. Accepted physical releases: 0.

# OpenSourceRail v0.5 — ferrite PM maglev

docs/open-source-rail-overview.md

Fewer moving parts, local manufacture and compact elevated urban routes.




OpenSourceRail maglev station concept

The primary design uses fixed onboard ferrite magnetic modules and powered linear electromagnetic propulsion, replacing conventional wheelsets, axles, gearboxes and rotating traction motors in normal running gear. Windings, drives, cooling, sensors, protection, passenger doors and emergency systems remain necessary. Contactless running aims to reduce wheel/rail contact noise and wear; acoustic performance and whole-system maintenance still require measurement.

## Simple civil works and local capability

The research target is one compact integrated concrete guidebeam per direction, with necessary ferromagnetic reaction surfaces, repeatable supports and no continuous external walkway. A vertical spiral evacuation chute is the preferred alternative under investigation. Walkway removal has not been approved. Route clearance, safe landings, assisted evacuation, stopping/recovery and fire response must work at every permitted stopping location.

The 99% elevated-mainline objective and straight tangents aim to reach constrained urban corridors with less ground occupation and disruption. Land rights, utilities, access, wildlife, waterways, curvature and foundation feasibility still govern each route. No universal clearance or zero-disruption claim is made.

Local casting, steel fabrication, cartridge assembly, winding, vehicle structures, interiors, wiring, Rust software and integration can create domestic capability. The complete local-content percentage, installed programme cost and funding requirement remain unknown until origin-traced bills, processes and supplier offers exist. Costs use the current material quantities, manufacturing routes and explicit planning rates. Unpriced scope remains open; no supplier quotation is implied.

# Inspect and reproduce the current work

| Current digital capability | Scope |
|---|---|
| City planning | 266 separate primary maglev packages; retained geography/demand, recalculated formation, duty, fleet and charging |
| Magnetic interface | 624 native solids; partial lift/cartridge/frame geometry |
| Beam-only civil kit | 54 native solids; selected geometry without external walkways |
| Evacuation entry and support | 20 partial entry parts and 3 support envelopes; not complete manufactured products |
| Controls | Allocation-free Rust core, 24-module replay, I/O contracts and advisory evacuation state machine |
| Simulation and media | Field equation controls, actual two-way Rust/beam replay, five assembly visibility videos, native images and source receipts |
| Engineering acceptance | 0 accepted physical releases; design, manufacture and operational gates remain open |

Use `./osr build` to verify dependencies and rebuild stale implemented studies. `./osr workbench` opens the primary maglev UI. `./osr design check` independently regenerates the report. `./osr book` builds the current reader edition. The scheduled runner repeats dependency verification after source changes; it does not invent missing physical tests or certify engineering completion.

Open the current Workbench, review calculations, CAD and videos, read the implementation and open work, or download the primary reader book.

# OpenSourceRail v0.5 ferrite-maglev engineering

engineering/ferrite_maglev/README.md

The current design uses ferrite PM-biased active-attraction support, fixed onboard windings and electronics, passive reaction iron and compact concrete guidebeams. Rust control packages run offline models for eventual off-the-shelf SBC implementation. Powered linear propulsion, cooling, sensing, independent protection, braking, capture, parking and recovery remain necessary. Accepted engineering and physical releases: zero.

## Run and inspect

```
./osr build
./osr design run
./osr design iterate
./osr design check
./osr design verify
./osr city baghdad
./osr sim baghdad --cycles 2
./osr workbench
./osr test
./osr book
cargo test --locked -p osr-maglev-control --all-targets
```

Generated report · Workbench · city directory · architecture.

`./osr capabilities` publishes executable sources and their scope. `./osr catalogue plan --city baghdad` resolves a selected regeneration, and `./osr catalogue run --city baghdad` rebuilds that city. `./osr readiness` reports current-source gates. `./osr design release-check` remains held while acceptance is open.

The dependency runner verifies implemented native, numerical, control, city and publication prerequisites in order. It rebuilds stale evidence and holds dependent outputs after failures. An interprocess lock excludes concurrent timer/manual builds. Runtime progress is stored under `build/maglev-workflow/latest.json`. This is a finite reproducible build of implemented evaluators; missing physical tests and engineering decisions require additional work.

The adaptive search, configured in iteration.toml, generates new section/span, vehicle-mass allocation and actual Rust-gain proposals from preceding results. Each viable proposal faces three coupled load/disturbance cases and replay/excess-gap controls. Shortlisted exact revisions face two separate confirmation scenarios. Failed confirmations remove recommendation eligibility while preserving analytical history. The seeded campaign has explicit evaluation and stagnation budgets; exact-source cache receipts support restart. A timer resumes stale campaigns, then idles once current.

Published campaign includes ancestry, traces, progress and the research recommendation. Preference weights and screening bounds are provisional. Scaled body/equipment mass and inertia assumptions do not replace complete manufactured geometry, magnetic measurements or qualified dynamics. The recommendation is kept separate from the reference configuration and city costing.

## Current inputs and design records

| Concern | Source |
|---|---|
| Architecture | Infrastructure policy |
| Vehicle and civil configuration | Reference configuration, configuration reader, study assumptions |
| Requirements and work allocation | Requirements data, current systems register |
| Cost quantities and manufacturing | Process model, planning rates |
| Magnetic/native geometry | Geometry, field equations, native export |
| City service and geography | Each city's `city.json`; compiler |
| Control | Rust maglev package, offline SBC profile |

Baghdad uses six 17 m bodies with shared coupling, platform, magnetic-arm and boarding-gap assumptions. The partial station model includes floor frames, guidebeams, island platform and a centred flush lift floor. It omits complete vehicle and station equipment; the boarding gap requires a qualified bridge plate. Native exports identify the exact modeled revision.

# Implementation and closure status

| Capability | Implemented scope | Outstanding evidence |
|---|---|---|
| Systems register | Current system items, requirement relationships, work packages, manufacturing and test obligations | Approved quantitative criteria and accepted exact-configuration evidence |
| Native geometry | Partial magnetic interface, civil bay, beam-only variant, evacuation entry, support envelopes and Baghdad station; FreeCAD/STEP exports, reopening checks, images and short visibility videos | Full production geometry, qualified materials, joints, loads, tolerance chains and manufacturing drawings |
| Magnetic field | 2D variable-reluctivity equation, manufactured-solution controls and explicit nonlinear material hypotheses | Coupon force convergence, measured material data, 3D force/loss maps and physical rig correlation |
| Rust control | Module allocation, five-mode virtual replay, protection interfaces, support-preserving stops and advisory evacuation | Actual actuator/sensor model, isolated physical I/O, independent protection and measured timing |
| Coupled replay | Two-way exchange of Rust module forces and flexible vertical-beam deflections | Flexible vehicle, lateral/torsional/joint/soil dynamics, moving loads and correlated stability/ride/fault measurements |
| Adaptive design | Result-guided vehicle/beam/gain variation, actual Rust coupling, protection controls, Pareto observations and separate confirmation | Full production geometry, qualified criteria, measured plant, whole-train dynamics and independent acceptance |
| City planning | Capacity-matched formations, duty, fleet, charging, guideway quantities, provisional costs and manufacturing process needs | Located terminal loops, route/ground/utility surveys, recovery, full duty and complete programme economics |
| Industrialisation | Current material quantities and explicit process/rate assumptions | Configured suppliers, qualified processes, origin-traced MBOM, production capacity, logistics and installed scope |

Assembly videos show final-position visibility sequences. They do not simulate motion, collision-free erection, crane operation or chute deployment. CAD export integrity does not establish strength or manufacturing readiness. Virtual replay steps do not demonstrate physical deadlines. Equation residuals do not establish real lift capacity.

# Manufacturing and costs

Cost estimates start from current maglev quantities and fabrication operations. The reference formation is six cars. Per-car equipment, shared train equipment, beam/support quantities, station and factory scope must be kept distinct. Planning rates are explicit assumptions, rather than supplier quotations. A missing first-pilot/NRE total stays unpriced. Project scope, contingency, lifecycle and finance require their own evidence.

City `engineering/maglev/manufacturing.json` records the materials, processing, equipment needs and construction approaches behind its estimate. Candidate processes include controlled precast casting/curing and inspection, reinforcement/prestress work, reaction-iron fabrication and fitting, structural cutting/joining, coil winding, cartridge assembly, wiring, software integration and factory testing. Transport, temporary works, crane radius/capacity, joints and site acceptance remain part of installed delivery.

Prefer verified British manufacture when considering UK suppliers. A UK distributor or integrator does not establish UK manufacturing origin. Likewise domestic assembly does not change imported cell, semiconductor, sensor or raw-material origin. Complete local-content percentage remains unknown until the complete bill and origin evidence exist.

# Evacuation and compact civil works

The research target is one integrated beam per direction without a continuous external walkway. Necessary reaction iron, supports, foundations and connections remain. A vertical spiral controlled-descent chute is the preferred unqualified onboard candidate. The evacuation study tracks variable height, anchorage, ground landing, assistance, fire/weather, partial failure, rescue, lifecycle and driverless operation. Walkway removal is conditional on accepted route and onboard alternatives at every permitted stopping location.

# Implementation roadmap

The closure queue covers complete mechanical/civil geometry, magnetic qualification, dynamics, physical controls, origin-traced industrialisation, city site evidence and demonstrator acceptance. Requirements allocation and successful digital checks define progress within their scope; neither constitutes a manufacturing or operating release.

# osr-maglev-control

crates/osr-maglev-control/README.md

Portable v0.5 maglev study core, using Rust `no_std`, no heap and no external dependencies. It targets simulation and off-the-shelf SBC replay. Every computed output has `physical_output_authorised = false`; the I/O boundary records study outputs and supplies no current, gate-drive or PWM operation.

The existing `evaluate(previous, inputs, now_ns, params)` API remains a single-module integer current hypothesis. Its polarity, damping, envelope, time/replay, stop/reset and support-power tests remain compatible. Increasing an upward-attraction gap asks for increasing support. The controller cannot be reused unchanged for shaped-pole passive vertical support.

## 24-module force and moment study

`array::evaluate_array` adds coupled heave, roll, pitch, lateral displacement and yaw feedback and `array::allocate` distributes requested forces and moments over 24 stations. This is one rigid car, not the complete six-car Baghdad train or a validated six-degree-of-freedom suspension. Longitudinal force has no actuator in this model; a nonzero longitudinal demand appears explicitly as unachieved residual. Propulsion, braking, roll/yaw cross-coupling from real traction, flexible car/beam modes, nonlinear finite-angle dynamics and contact dynamics remain open.

Coordinates use x forward, y left and z up, with the assumed car centre of mass as origin. Station positions and gaps are metres, force is newtons, moments are newton-metres, small angles are radians and timestamps are monotonic nanoseconds. Upward force is positive attraction. Lateral force is positive left. Each moment is the sum of `position cross force`. The explicit seed geometry is 12 paired stations at x = -5.5, -4.5, …, 5.5 m and y = ±0.88 m, z = 0 m; this matches the candidate 12 m magnetic strips. Actual centre of mass, attachment heights, lever arms, tolerances and dynamic mass properties need physical closure.

Each station has a bounded upward channel and a signed lateral channel. The illustrative limits are 0–20 kN upward, ±4 kN lateral and 2 MN/s slew per channel. These numbers are hypothetical force envelopes, not measured magnetic maps, current/voltage/thermal limits or proved simultaneous lift/guidance capability. A real PM-biased module can retain force after drive loss: a failed station takes an explicit specified residual force. Zero in the example is an ideal failure assumption. Unknown failed-channel forces require a separate bounded uncertainty and protection model before hardware use.

The allocator uses at most 256 bounded coordinate sweeps to minimise squared force residuals and moment residuals scaled by a declared 6 m lever arm. It reports the independently reconstructed resultant, residual wrench, constraint limits and feasibility tolerances. A solver iteration count establishes no physical deadline. Infeasible requests remain visible; no six-axis success is inferred from solver completion. Slew limits apply against a prior force frame; without one the first study allocation has no force history and applies only static limits. This is not a physical startup or touchdown procedure.

The mode estimator fits a small-angle rigid plane to 24 dual-gap samples and fits lateral displacement/yaw to 24 dual lateral samples. Every individual gap and lateral sample is checked before averaging; disagreement, nonfinite values and degenerate geometry fail. This fit is not a flexible-body observer. Sensor channels share one supplied acquisition timestamp; per-channel skew, physical independence, common-cause failure and calibration are not established. Derivatives use acquisition time. Host time/sequence checks reject stale, future, replayed, regressing and late frames.

A stop request preserves fresh bounded support. A station failure, active bound or infeasible allocation latches a traction stop while available ideal support remains active. Missing valid sensing or power requests external local fallback, never frozen study forces. A prior fault inhibits traction until a fresh healthy frame, reported stopped/secured conditions and reset coincide. Capture, fallback, independent protection energy and secure-condition verification need separate engineering; they are not supplied by these Boolean reports.

## Reproducible offline checks

Run:

```
cargo test -p osr-maglev-control
cargo clippy -p osr-maglev-control --all-targets -- -D warnings
cargo run -p osr-maglev-control --example sbc_replay
cargo run -p osr-maglev-control --example multimodule_replay
```

The array replay integrates ten seconds of virtual 1 ms frames, prints CSV at 500 ms intervals and asserts feasibility and a final small five-mode error. Its 27,750 kg AW2 mass is the profile's 18,750 kg matched tare plus 120 × 75 kg passengers. Gains, diagonal inertias, force limits and instantaneous actuator response are synthetic. It starts with coupled displacement/tilt, commands a stop at 0.5 s and loses station 0 at 1 s under a specified zero-residual failure hypothesis. It keeps fresh support and the stop latch active. This trajectory is a numerical example, not a closed-loop stability or ride-quality

proof.

`ideal::IdealPlant` integrates a constant wrench exactly within each small-angle step. Tests independently check uniform equilibrium (272,227.5 N weight), moment signs, force/moment reconstruction, five-component demand, failed-force redistribution, capacity and slew infeasibility, all-failed residuals, sensor and time rejection, support-preserving stops and secure reset. Mechanical interface power agrees with resultant wrench power, and one-step work agrees with change in kinetic plus gravitational energy. These conservation checks validate the ideal mechanics arithmetic, not actual magnet energy or electrical losses.

`io::StudyIo`, `host_tick` and `ReplayIo` define an acquisition/record boundary without owning an OS clock or hardware. Recording/acquisition failures return to the host; the host must engineer its local fallback. Repeated single-frame stub reads are rejected as replay. No scheduler, GPIO, ADC or network transport is validated, and this example measures neither SBC timing nor I/O latency.

Physical gates remain open: calibrated nonlinear magnetic force/loss maps over gap/current/temperature and shared flux, identified electrical/thermal and mechanical dynamics, actuator failure residuals, independent sensors/protection, measured end-to-end target timing, multi-module rigs, flexible guideway and loaded vehicle tests, braking/capture and train/route acceptance. No operational or manufacturing authority is inherited from the numerical study.

# Flexible guideway numerical demonstrator

`coupled_bridge` connects this exact array core to the separately labelled Python beam demonstrator in `engineering/ferrite_maglev/coupled.py`. From the repository root, run:

```
cargo build -p osr-maglev-control --release --example coupled_bridge
.venv/bin/python -m engineering.ferrite_maglev.coupled --duration-s 3
.venv/bin/python -m pytest -q engineering/analysis/tests/test_ferrite_maglev_coupled.py
```

The Python host sends all 24 local relative gaps each virtual millisecond. Rust returns bounded support forces, which load a 25 m simply supported Euler–Bernoulli beam; its local displacement feeds the next sensor frame. This is two-way offline coupling, with no physical output. The beam assumes the current configured section and native quantity-derived line mass, with 2% modal damping, retaining six vertical modes of a 50-element consistent-mass model. Independent point-load closed forms check static response; analytical pinned-beam frequencies and virtual-work/undamped-energy checks control the numerical implementation.

The model can show global beam displacement exceeding 8 mm while the car follows that displacement and the relative gaps remain inside the example's study envelope. This establishes neither a gap requirement nor structural acceptance. It omits dead-load camber, torsion/corbel flexibility, moving/train loads, flexible car structure, soil, joints, fatigue, real magnetic/electrical dynamics and independent protection. The line protocol accepts at most 20,000 frames, 8 KiB per frame, and measures no hardware timing. The three-mode car motion and vertical beam coupling do not provide a six-degree-of-freedom stability proof.

# Offline evacuation sequencing

`egress::evaluate` adds allocation-free doorway sequencing with an exact declared candidate/door context, fresh monotonic frames, explicit readiness/assistance prerequisites, latched contradiction/context/deadline faults, and secured/stowed reset. It reports study traction inhibition and continued-support obligations; it supplies no door, restraint, chute or support output adapter. Input assertions are unauthenticated study facts, not verified independent sensor or safety evidence.

`cargo run -p osr-maglev-control --release --example egress_replay -- REVISION PATH_CLEAR HEIGHT_COVERED` prints a bounded nine-frame synthetic sequence. The linked Python study supplies its candidate tag and calculated hypothetical path/height gates, and retains negative controls. Loss of prerequisites after deployment begins holds the sequence; no automatic retraction is commanded. The egress/support composition test preserves fresh support control while inhibiting traction. Physical timing, independent protection, accessible passenger evacuation and railway approval remain open.

# Adaptive research tuning

`./osr design iterate` varies bounded proportional and damping gains alongside vehicle mass allocations and guidebeam sections/spans. The bridge accepts mass and gain scales as optional command arguments; nonfinite or out-of-bound scales are rejected before processing frames. Exact revisions face separate confirmation scenarios and replay/excess-gap negative controls. This offline tuning establishes neither measured magnetic response nor stability over the full operating envelope.

# OpenSourceRail architecture

docs/ARCHITECTURE.md

OpenSourceRail v0.5 develops ferrite PM-biased active-attraction maglev for urban transport. Its current physical basis is fixed onboard magnetic modules, powered linear propulsion, passive reaction iron and compact elevated concrete guidebeams. Batteries and station charging support operation; independent protection, cooling, capture, braking, parking and recovery remain essential design obligations.

The objective is broad city access with a small ground footprint, few moving running-gear parts and practical local manufacture. Ordinary mainlines target repeated elevated bays. Route rights, foundations, utilities, curvature, station access and evacuation determine where the concept is feasible.

## Linked design layers

| Layer | Controlled definition | Evidence |
|---|---|---|
| Service and city | Geography, demand, capacity, headways, route lengths, station locations and local assumptions | Formation, fleet, energy/charging, quantities, provisional costs and unresolved route gates |
| Mechanical | Magnetic cartridges, structural frame, passenger systems, energy, doors and evacuation | Partial native solids, mass budgets, interface checks, analytical screens and production obligations |
| Civil | Guidebeams, reaction iron, joints, supports, foundations, stations and services | Section/beam calculations, support-demand studies, native envelopes and manufacturing/erection quantities |
| Software | Rust support control, protection interfaces, module allocation, fault handling and advisory evacuation | Offline executable tests, virtual replay and a two-way flexible-beam coupling |
| Industrialisation | Materials, fabrication processes, tooling, factory throughput, transport, erection and origin | Quantity-and-process planning allowances and explicit unpriced scope |
| Acceptance | Exact revision, requirements, tests, evidence, reviewer and serial identity | Source-bound receipts and open engineering/manufacturing/physical gates |

Infrastructure policy, configuration and engineering modules control these interfaces. Generated city inputs use `city.json`; their outputs follow the same maglev compiler and cost model.

## Current evidence boundary

Native CAD covers selected partial assemblies. Assembly visibility videos reveal final-position geometry; they do not establish collision-free installation or deployment. Field-equation controls do not establish real magnetic lift capacity. Virtual control/beam replay does not establish physical SBC deadlines or qualified train dynamics. Open geographic data supports planning estimates, rather than surveyed boundaries, soil capacity or utility clearance.

A continuous-walkway-free guideway remains conditional on qualified onboard and route evacuation. Vertical spiral chutes require usable landings and assisted evacuation at every allowed stopping location. No blanket waiver of evacuation or structural requirements is implied by the compact layout.

All release flags remain false and accepted engineering/physical releases remain zero. Local content, installed programme cost and the funding requirement require complete origin-traced quantities, configured supplier offers, construction scope and financing terms. Successful publication integrity is separate from engineering acceptance.

# Roadmap

docs/ROADMAP.md

OpenSourceRail v0.5 uses one ferrite-maglev architecture across mechanical, civil, software, manufacturing and city planning. Implemented work and commands and the current report identify the tested digital scope. Accepted engineering and physical releases remain zero.

## Primary maglev closure queue

| Package | Evidence required for closure |
|---|---|
| Requirements and configuration | Approved loads, service envelope, tolerances and quantitative acceptance criteria for exact revisions |
| Magnetic assemblies | Complete support/guidance/propulsion/capture assemblies, connections and dimensional chains |
| Magnetic force and loss | Converged force calculations, qualified material curves, 3D winding/end effects, thermal maps and correlated rig measurements |
| Vehicle | Complete body, doors, interiors, storage, cooling, controls and evacuation; mass, centre of gravity, inertia and structural qualification |
| Civil kit | Beams, reaction iron, connections, supports, foundations, stations, services and all handling/construction stages |
| Control and protection | Qualified plant model, independent protection, loss-of-power behavior, braking, capture and recovery interfaces |
| Coupled dynamics | Moving trains, flexible bodies, lateral/torsional modes, joints and soil with calibrated models |
| SBC hardware | Isolated I/O, power stages, measured deadlines, watchdogs and independent protection on selected equipment |
| City delivery | Located routes, terminal loops, platforms, charging, storage/recovery, all-stop evacuation and site investigations |
| Manufacturing and economics | Complete origin-traced MBOM, process routes, machine/tooling options, throughput, transport, erection, maintenance and financing |
| Demonstrator | Traceable manufactured units, independent inspections and physical acceptance on identified serials and revisions |

The adaptive designer now varies vehicle mass allocations, beam geometry/spans and actual Rust controller gains using previous measured results. It retains ancestry and traces, applies separate holdout scenarios and disqualifies failed confirmations. Its preferred candidate is a bounded research result; it does not close production geometry, qualified magnetic maps, physical timing or whole-train acceptance.

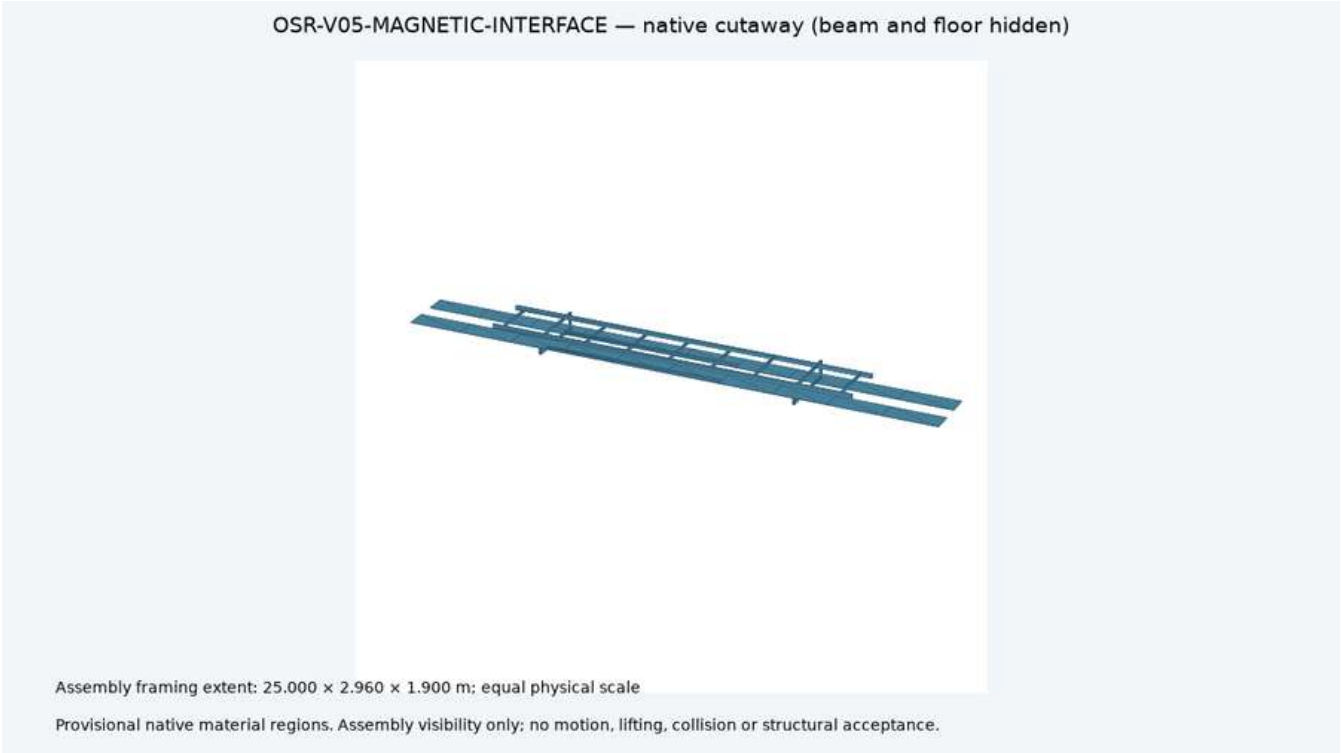
The automated runner can regenerate the implemented studies and hold downstream evidence after failures. Missing evaluators, measurements and approvals remain open work, even after every current automated check passes.

# Native geometry review




Partial six-car Baghdad station, magnetic guidebeams and island lift; acceptance remains open.

Partial six-car Baghdad station, magnetic guidebeams and island lift; acceptance remains open.




Partial fixed magnetic interface; not a complete vehicle.

Partial fixed magnetic interface; not a complete vehicle.



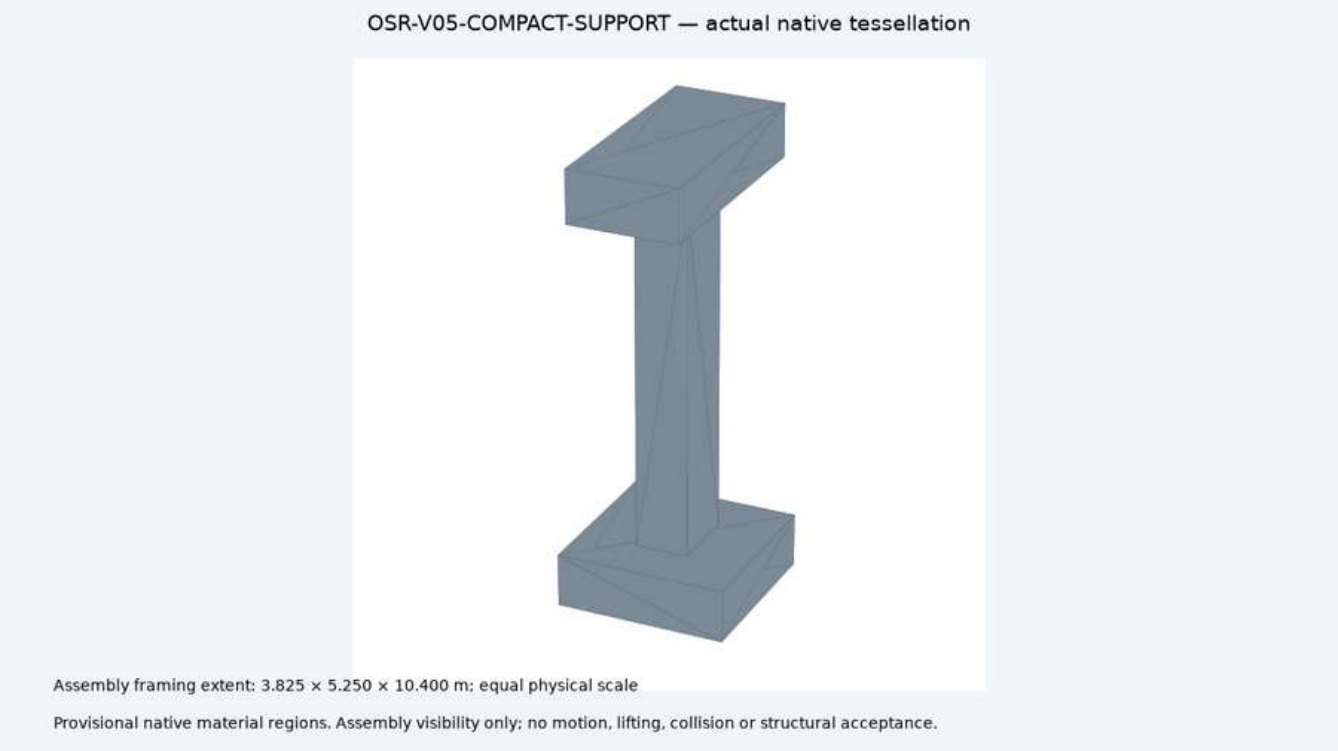

Beam-only civil variant. Supports are envelopes; evacuation acceptance remains open.

Beam-only civil variant. Supports are envelopes; evacuation acceptance remains open.




Partial evacuation entry frame and empty cassette. No deployed or stowed mechanism is qualified.

Partial evacuation entry frame and empty cassette. No deployed or stowed mechanism is qualified.




Compact pier and foundation envelope. Site capacity and structural acceptance remain open.

Compact pier and foundation envelope. Site capacity and structural acceptance remain open.

# Primary maglev city directory

Separate planning packages. Complete programme costs, local content, land and physical acceptance remain open for every city.

| City | Provisional trainsets | Cars | Open gates |
|---|---|---|---|
| Baghdad | 550 | 3300 | 12 |
| Benguela | 115 | 345 | 12 |
| Huambo | 91 | 273 | 12 |
| Lobito | 64 | 192 | 12 |
| Luanda | 650 | 3900 | 12 |
| Lubango | 119 | 357 | 12 |
| Malanje | 28 | 84 | 12 |
| Namibe | 79 | 158 | 12 |
| Soyo | 63 | 126 | 12 |
| Uige | 17 | 51 | 12 |
| Bukavu | 118 | 354 | 12 |
| Goma | 85 | 255 | 12 |
| Kananga | 50 | 200 | 12 |
| Kinshasa | 1334 | 8004 | 12 |
| Kisangani | 64 | 256 | 12 |
| Lubumbashi | 192 | 768 | 12 |
| Mbuji Mayi | 132 | 528 | 12 |
| Eldoret | 133 | 399 | 12 |
| Garissa | 48 | 96 | 12 |
| Kakamega | 67 | 134 | 12 |
| Kisii | 50 | 100 | 12 |
| Kisumu | 79 | 237 | 12 |
| Kitale | 66 | 132 | 12 |
| Machakos | 64 | 128 | 12 |
| Malindi | 53 | 106 | 12 |
| Meru Ke | 64 | 128 | 12 |
| Mombasa | 359 | 1436 | 12 |
| Nairobi | 822 | 4932 | 12 |
| Naivasha | 86 | 172 | 12 |
| Nakuru | 84 | 252 | 12 |
| Nyeri | 57 | 114 | 12 |
| Thika | 98 | 294 | 12 |
| Antananarivo | 536 | 3216 | 12 |
| Beira | 69 | 207 | 12 |
| Chimoio | 54 | 162 | 12 |
| Lichinga | 28 | 56 | 12 |
| Maputo | 249 | 996 | 12 |
| Nacala | 77 | 154 | 12 |
| Nampula | 80 | 240 | 12 |
| Pemba Mz | 66 | 132 | 12 |
| Quelimane | 15 | 45 | 12 |
| Tete | 88 | 264 | 12 |
| Xai Xai | 40 | 80 | 12 |
| Huye | 73 | 146 | 12 |
| Kigali | 204 | 816 | 12 |
| Rubavu | 61 | 122 | 12 |
| Mogadishu | 232 | 928 | 12 |
| Waw | 45 | 90 | 12 |
| Arusha | 87 | 261 | 12 |
| Dar Es Salaam | 800 | 4800 | 12 |
| Dodoma | 80 | 240 | 12 |
| Iringa | 64 | 128 | 12 |
| Kigoma | 52 | 104 | 12 |
| Mbeya | 77 | 231 | 12 |
| Morogoro | 76 | 228 | 12 |
| Moshi | 57 | 114 | 12 |
| Mwanza | 251 | 1004 | 12 |

| City | Provisional trainsets | Cars | Open gates |
|---|---|---|---|
| Shinyanga | 90 | 180 | 12 |
| Songea | 19 | 38 | 12 |
| Sumbawanga | 35 | 70 | 12 |
| Tabora | 47 | 94 | 12 |
| Tanga | 83 | 249 | 12 |
| Zanzibar City | 83 | 249 | 12 |
| Arua | 67 | 134 | 12 |
| Entebbe | 78 | 156 | 12 |
| Fort Portal | 55 | 110 | 12 |
| Gulu | 129 | 387 | 12 |
| Hoima | 53 | 106 | 12 |
| Jinja | 92 | 184 | 12 |
| Kampala | 300 | 1200 | 12 |
| Lira | 88 | 176 | 12 |
| Masaka | 62 | 124 | 12 |
| Mbale | 64 | 128 | 12 |
| Mbarara | 89 | 267 | 12 |
| Soroti | 21 | 42 | 12 |
| Lusaka | 337 | 2022 | 12 |
| Lyon | 357 | 1428 | 12 |
| La Paz | 252 | 1008 | 12 |
| Cuenca | 101 | 303 | 12 |
| San Salvador | 294 | 1176 | 12 |
| Agadir | 120 | 360 | 12 |
| Beni Mellal | 46 | 92 | 12 |
| Fez | 191 | 764 | 12 |
| Kenitra | 105 | 315 | 12 |
| Khouribga | 39 | 78 | 12 |
| Marrakech | 322 | 1288 | 12 |
| Meknes | 82 | 246 | 12 |
| Nador | 67 | 134 | 12 |
| Oujda | 56 | 168 | 12 |
| Safi | 64 | 192 | 12 |
| Tangier | 247 | 988 | 12 |
| Tetouan | 98 | 294 | 12 |
| El Obeid | 81 | 243 | 12 |
| Kassala | 41 | 123 | 12 |
| Khartoum | 550 | 3300 | 12 |
| Nyala | 72 | 216 | 12 |
| Omdurman | 441 | 1764 | 12 |
| Port Sudan | 61 | 183 | 12 |
| Tunis | 275 | 1100 | 12 |
| Bloemfontein | 190 | 570 | 12 |
| Durban | 586 | 3516 | 12 |
| East London Za | 91 | 273 | 12 |
| Nelspruit | 57 | 114 | 12 |
| Polokwane | 83 | 249 | 12 |
| Herat | 64 | 192 | 12 |
| Jalalabad Af | 77 | 231 | 12 |
| Kabul | 466 | 2796 | 12 |
| Kandahar | 88 | 264 | 12 |
| Mazar E Sharif | 111 | 333 | 12 |
| Barisal | 108 | 324 | 12 |
| Chittagong | 519 | 3114 | 12 |
| Comilla | 150 | 450 | 12 |
| Gazipur | 311 | 1244 | 12 |
| Khulna | 255 | 1020 | 12 |
| Mymensingh | 85 | 255 | 12 |
| Narayanganj | 123 | 369 | 12 |
| Rajshahi | 112 | 336 | 12 |
| Rangpur | 70 | 210 | 12 |
| Sylhet | 81 | 243 | 12 |
| Agra | 159 | 636 | 12 |

| City | Provisional trainsets | Cars | Open gates |
|---|---|---|---|
| Bhopal | 235 | 940 | 12 |
| Coimbatore | 435 | 2610 | 12 |
| Indore | 412 | 2472 | 12 |
| Jodhpur | 176 | 704 | 12 |
| Kanpur | 378 | 2268 | 12 |
| Lucknow | 371 | 2226 | 12 |
| Madurai | 243 | 972 | 12 |
| Meerut | 154 | 616 | 12 |
| Patna | 398 | 1592 | 12 |
| Raipur | 154 | 616 | 12 |
| Rajkot | 133 | 532 | 12 |
| Ranchi | 283 | 1132 | 12 |
| Vadodara | 198 | 792 | 12 |
| Varanasi | 222 | 888 | 12 |
| Vijayawada | 206 | 824 | 12 |
| Visakhapatnam | 264 | 1056 | 12 |
| Biratnagar | 55 | 110 | 12 |
| Kathmandu | 373 | 1492 | 12 |
| Pokhara | 126 | 378 | 12 |
| Bahawalpur | 79 | 237 | 12 |
| Faisalabad | 235 | 1410 | 12 |
| Gujranwala | 207 | 828 | 12 |
| Hyderabad Pk | 296 | 1184 | 12 |
| Karachi | 689 | 4134 | 12 |
| Larkana | 50 | 150 | 12 |
| Multan | 164 | 656 | 12 |
| Peshawar | 204 | 816 | 12 |
| Quetta | 125 | 500 | 12 |
| Rahim Yar Khan | 99 | 297 | 12 |
| Sheikhupura | 58 | 174 | 12 |
| Sialkot | 83 | 249 | 12 |
| Sukkur | 73 | 219 | 12 |
| Colombo | 473 | 2838 | 12 |
| Galle | 99 | 297 | 12 |
| Jaffna | 95 | 285 | 12 |
| Kandy | 107 | 321 | 12 |
| Phnom Penh | 432 | 1728 | 12 |
| Bandung | 419 | 1676 | 12 |
| Surabaya | 469 | 2814 | 12 |
| Vientiane | 117 | 351 | 12 |
| Mandalay | 266 | 1064 | 12 |
| Yangon | 691 | 4146 | 12 |
| Davao | 309 | 1236 | 12 |
| Ouagadougou | 294 | 1176 | 12 |
| Bafoussam | 95 | 285 | 12 |
| Bamenda | 80 | 240 | 12 |
| Bertoua | 36 | 108 | 12 |
| Douala | 243 | 1458 | 12 |
| Edea | 11 | 22 | 12 |
| Garoua | 59 | 177 | 12 |
| Kumba | 62 | 186 | 12 |
| Maroua | 80 | 240 | 12 |
| Ngaoundere | 44 | 132 | 12 |
| Yaounde | 334 | 2004 | 12 |
| Conakry | 139 | 556 | 12 |
| Bamako | 413 | 1652 | 12 |
| Niamey | 233 | 932 | 12 |
| Aba Ng | 56 | 168 | 12 |
| Benin City | 230 | 920 | 12 |
| Ibadan | 259 | 1554 | 12 |
| Ilorin | 79 | 237 | 12 |
| Jos | 91 | 273 | 12 |
| Kano | 559 | 3354 | 12 |

| City | Provisional trainsets | Cars | Open gates |
|---|---|---|---|
| Maiduguri | 209 | 836 | 12 |
| Onitsha | 231 | 924 | 12 |
| Port Harcourt | 231 | 924 | 12 |
| Uyo | 63 | 189 | 12 |
| Dakar | 372 | 2232 | 12 |
| Arish | 24 | 48 | 12 |
| Asyut | 80 | 240 | 12 |
| Beni Suef | 55 | 165 | 12 |
| Damanhur | 82 | 246 | 12 |
| Damietta | 110 | 330 | 12 |
| Fayoum | 92 | 276 | 12 |
| Hurghada | 54 | 108 | 12 |
| Ismailia | 90 | 270 | 12 |
| Kafr El Sheikh | 60 | 120 | 12 |
| Luxor | 103 | 309 | 12 |
| Mahalla | 58 | 174 | 12 |
| Mansoura Eg | 75 | 225 | 12 |
| Minya | 89 | 267 | 12 |
| Port Said | 54 | 162 | 12 |
| Qena | 95 | 285 | 12 |
| Sohag | 84 | 252 | 12 |
| Suez | 89 | 267 | 12 |
| Tanta | 107 | 321 | 12 |
| Zagazig | 64 | 192 | 12 |
| Amarah | 76 | 228 | 12 |
| Baqubah | 110 | 330 | 12 |
| Basra | 399 | 2394 | 12 |
| Diwaniyah | 88 | 264 | 12 |
| Duhok | 86 | 258 | 12 |
| Erbil | 202 | 808 | 12 |
| Fallujah | 80 | 240 | 12 |
| Hillah | 95 | 285 | 12 |
| Karbala | 208 | 832 | 12 |
| Kirkuk | 197 | 788 | 12 |
| Kut | 70 | 210 | 12 |
| Mosul | 257 | 1028 | 12 |
| Najaf | 246 | 984 | 12 |
| Nasiriyah | 79 | 237 | 12 |
| Ramadi | 85 | 255 | 12 |
| Samawah | 84 | 252 | 12 |
| Sulaymaniyah | 121 | 484 | 12 |
| Amman | 440 | 2640 | 12 |
| Aqaba | 55 | 110 | 12 |
| Irbid | 76 | 228 | 12 |
| Zarqa | 103 | 309 | 12 |
| Beirut | 214 | 856 | 12 |
| Sidon | 54 | 108 | 12 |
| Tripoli Lb | 72 | 216 | 12 |
| Gaza City | 47 | 141 | 12 |
| Hebron | 108 | 324 | 12 |
| Nablus | 109 | 327 | 12 |
| Abha | 99 | 297 | 12 |
| Al Kharj | 98 | 294 | 12 |
| Buraidah | 100 | 300 | 12 |
| Dammam | 313 | 1252 | 12 |
| Hail | 116 | 348 | 12 |
| Hofuf | 111 | 333 | 12 |
| Jeddah | 491 | 2946 | 12 |
| Jizan | 90 | 270 | 12 |
| Khamis Mushait | 148 | 444 | 12 |
| Mecca | 252 | 1008 | 12 |
| Medina | 208 | 832 | 12 |
| Najran | 88 | 264 | 12 |

| City | Provisional trainsets | Cars | Open gates |
|---|---|---|---|
| Tabuk | 96 | 288 | 12 |
| Taif | 93 | 279 | 12 |
| Aleppo | 383 | 1532 | 12 |
| Damascus | 294 | 1176 | 12 |
| Deir Ez Zor | 90 | 270 | 12 |
| Hama | 111 | 333 | 12 |
| Homs | 68 | 204 | 12 |
| Idlib | 63 | 126 | 12 |
| Latakia | 90 | 270 | 12 |
| Raqqa | 69 | 207 | 12 |
| Tartus | 58 | 116 | 12 |
| Aden | 81 | 243 | 12 |
| Dhamar | 58 | 116 | 12 |
| Hodeidah | 48 | 144 | 12 |
| Ibb | 96 | 288 | 12 |
| Lahij | 49 | 98 | 12 |
| Mukalla | 122 | 366 | 12 |
| Sanaa | 405 | 2430 | 12 |
| Sayun | 42 | 84 | 12 |
| Taiz | 89 | 267 | 12 |